

When More Becomes Less: A Two-Probe Diagnostic for Repetition in Language Models

Anonymous ACL submission

Abstract

Many probing studies vary how often a target token appears in a prompt and read the model’s prediction at a single slot. The implicit assumption is that the slot’s position is orthogonal to how repetition affects prediction. We show this assumption fails systematically. Our two-probe diagnostic holds a repeated-target prefix fixed and varies only the readout position: the *adjacent* probe places the slot immediately after the repeated block; the *displaced* probe places it inside a fresh sentence frame. Adjacent repetition behaves as priming intuition predicts — $P(\text{target})$ climbs with N and plateaus. Displaced repetition produces an inverted-U: $P(\text{target})$ rises to an early peak and then declines as more copies are added. The displaced inverted-U shows a per-word drop with bootstrap CI excluding zero in all 13 open-access encoder and decoder models we test, with 12 of 13 mean curves also turning over before $N=30$. A six-condition causal ablation isolates the effect to exact lexical repetition rather than generic redundancy or semantic-neighbour exposure, and the same inverted-U replicates across Spanish, Chinese, German, and French in 42 of 42 multilingual cells. Internally, per-token attention to each repeated copy falls with N while the total budget the model assigns to the block grows in the three CLMs we probe but not in the one MLM. The methodological consequence: cloze-style probes that vary how often a target appears cannot treat the readout position as orthogonal to what they measure.

1 Introduction

A common pattern in probing studies of language models (Petroni et al., 2019; Ettinger, 2020) is to vary the number of times a target token appears in a prompt and to read the model’s prediction at a single slot — typically the next-token position after the variation. The implicit assumption is that the readout position is orthogonal to what repetition does: that whatever effect more copies have on

prediction is the same effect regardless of where the prediction slot sits relative to the repeated block. We show this assumption fails. Hold the repeated-target prefix fixed and the same prefix produces opposite curves at two readout positions: priming plateaus adjacent to the block, retrieval reverses one clause later.

The diagnostic is a minimal pair (Figure 1). We hold a repeated-target prefix fixed and vary only where the prediction slot lies. The *adjacent* probe places the slot directly after the block; the *displaced* probe moves it into a fresh sentence frame. Adjacent priming plateaus. Displaced retrieval reverses. The dissociation is what lets us read the displaced effect as non-additive aggregation rather than as long-context capacity loss — the adjacent baseline confirms local continuation is still available under the same prefix.

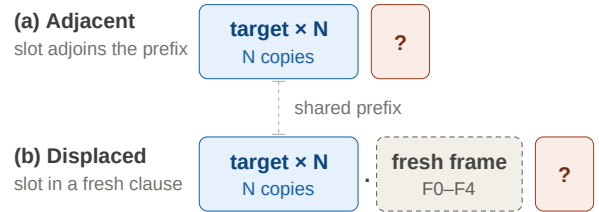


Figure 1: **Two-probe design.** (a) The *adjacent* probe places the prediction slot directly after a prefix of N target copies. (b) The *displaced* probe shares the same prefix but moves the slot into a fresh clause via one of five sentence frames F0–F4. The framework applies identically to masked LMs (slot is [MASK]) and to causal LMs (slot is the next-token position).

All 13 models show positive per-word drops with CI excluding zero (Table 1); 12 of 13 also show mean-curve inversion before $N=30$. SmolLM2-1.7B is the only checkpoint whose population-mean curve continues to climb through $N=30$, while its per-word drops remain positive (0.138 [0.114, 0.164]) — the secondary observation fails but the headline statistic does not. The cleanest

within-thesis illustration is a truncation contrast: when the displaced peak sits near $N=3$, cutting the prefix from $N=30$ back down to $N=3$ raises target probability with the bootstrap interval clearing zero in two of four causal-ablation models (ModernBERT-large +0.034 [0.016, 0.056], rank 35 $\rightarrow$ 7; Qwen2.5-1.5B +0.059 [0.046, 0.072], rank 75 $\rightarrow$ 7). Qwen2.5-7B goes in the same direction with a CI that barely overlaps zero (+0.015, [-0.005, 0.035]).

A six-condition causal ablation pins the effect to *exact* lexical repetition: length-matched random fillers and unrelated-token repetition flatten $P(\text{target})$, and semantic-neighbour fillers activate the target’s semantic field (raising $\Sigma P(\text{synonyms})$ to 0.32 in ModernBERT-large) without lifting $P(\text{target})$ itself above roughly 0.03 (max 0.031 in ModernBERT-large). The same shape replicates on four multilingual models across Spanish, Chinese, German, and French. Internally, the per-target-token attention each repeated copy receives falls with N while the *total* attention budget the model assigns to the repeated block grows — the block as a whole tightens its hold on attention even as each individual copy gets less.

Our primary contribution is methodological: a matched-prefix two-probe design that separates local pattern continuation from displaced retrieval, applicable to any controlled probing study of repeated content. The empirical finding our diagnostic makes visible — that current open-access models fail to aggregate exact lexical repetition additively at displaced slots — is the second contribution, validated by a four-language replication, a six-condition specificity check, and an internal correlate in the attention-budget dynamics of three CLMs.

2 Related Work

Most prior work on repetition in language models has been about the model’s output: degenerate loops under greedy or beam decoding (Holtzman et al., 2020; Welleck et al., 2020; Xu et al., 2022), and the use of repeated demonstrations as a positive force in in-context learning (Min et al., 2022; Agarwal et al., 2024). We are asking about the input side instead — how the model’s predictions change when the prompt is repetitive.

Several lines bear on what repeated tokens do to Transformer-based language models on the input side. Barbero et al. (2024) prove a representational-

collapse result for repeated decoder-only inputs; Yona et al. (2025) characterize a “repeated-token divergence” phenomenon and link it to attention-sink dynamics; attention sinks themselves (Xiao et al., 2024; Gu et al., 2025) have been characterized as positions that accumulate disproportionate attention mass. Our block-budget growth shares the flavour of the attention-sink line, but appears under a different setup — a span at the source side under displaced readout, rather than initial tokens carried across all queries — and we stop short of claiming it is the same mechanism. None of this prior work asks our question, which is whether the *position of the readout slot relative to the repeated block* changes what repetition does to retrieval. The experimental object here is a behavioural dissociation between two probe positions under a matched prefix.

Other prompt-side phenomena include “lost in the middle” effects under long prompts (Liu et al., 2024), the many-shot prompt-sensitivity results that bedevil in-context-learning benchmarks (Lu et al., 2022; Sclar et al., 2024; Agarwal et al., 2024), and retrieval-augmented generation and open-domain QA pipelines that invite exact and near-exact repetition by concatenating passages sharing an entity, an answer string, or a source document (Lewis et al., 2020; Karpukhin et al., 2020; Izacard and Grave, 2021).

Methodologically our paper is closest to the cloze-probe tradition (Petroni et al., 2019; Tenney et al., 2019; Ettinger, 2020; Misra et al., 2020; Hu and Levy, 2023), but with a *displaced* slot that dissociates surface continuation from across-clause retrieval, and with the framework extended to causal LMs by replacing [MASK] with the next-token position. The repetition schedule in our prefix is conceptually inspired by the psycholinguistic literature on semantic satiation (Smith and Klein, 1990; Black, 2004; Tian and Huber, 2010; Pilotti et al., 1997), but we do not claim a human-fatigue mechanism.

3 Experimental Design

3.1 Two-probe framework

Both probes use the same N grid $N \in \{0, 1, 3, 5, 10, 20, 30\}$ and a shared repeated-target prefix; only the readout slot differs. ($N=0$ is the prior-only baseline; all reported statistics use $N \geq 1$.) The *adjacent* probe is <target $\times$ N> [MASK] . (MLM) or <target $\times$ N> (CLM

next-token slot). The *displaced* probe is $\langle \text{target} \times N \rangle \cdot \langle \text{frame} \rangle [\text{MASK}] \cdot (\text{MLM})$ or $\langle \text{target} \times N \rangle \cdot \langle \text{frame} \rangle$ (CLM). We test five English declarative-style frames F0–F4 (Appendix A). The adjacent probe is evaluated on a three-model baseline (ModernBERT-base, Qwen2.5-1.5B, Qwen2.5-7B) as a sanity check that local continuation remains available; the displaced grid is evaluated on every panel model. A decline at the displaced slot therefore cannot be reduced to local capacity loss; the adjacent baseline establishes that local continuation is intact.

A short vocabulary note. *Displaced retrieval* names the readout condition just defined; *adjacent retrieval* names the matched local-continuation control. The label *satiation-like* is used only for per-word curves whose shape clears the deterministic thresholds of Appendix B — it is a curve label, not a mechanistic claim. *Non-additive aggregation* describes the shape of $P(\text{target} | N)$ at the displaced slot, not anything about internal representations.

3.2 Targets and tokenization

The English target set is 261 hand-curated single-token noun candidates balanced over concreteness and Zipf frequency (Speer, 2022); per-tokenizer filtering yields per-model sets of 182–258 words and a shared core of ≈ 168 words that enters the headline regression. Construction details (cell balance, semantic-class breakdown, per-family overlap) are in Appendix A.

3.3 Open-access panel

We probe 13 open-access language models, all loadable without authentication. Two encoders use the masked-LM objective (ModernBERT-base, ModernBERT-large (Warner et al., 2025)) and eleven decoders use the causal-LM objective. The Qwen2.5 family (Qwen Team, 2025) provides a within-family scale ladder (0.5B, 1.5B, 3B, 7B, 14B, plus the 7B-Instruct variant). Cross-family checks include SmoLLM2 at 360M and 1.7B (Ben Allal et al., 2025), Phi-3.5-mini at 3.8B (Abdin et al., 2024), and OLMo-2 at 1B and 7B (OLMo Team, 2025). Hardware, dtype, and prompt-format details (including evaluating instruction-tuned variants on the raw prompt rather than the chat template) are in Appendix H.

3.4 Metrics and statistics

Our headline shape statistic is the per-word drop $(\text{peak}_w - P(N=30)_w) / \text{peak}_w$ with the peak taken

over $N \in \{1, 3, 5, 10, 20, 30\}$. We bootstrap the population mean at the word level. Alongside the per-word drop we run an OLS quadratic of $\log P(\text{target})$ on N , N^2 , Zipf frequency, model, frame, and word category, with SEs clustered by target word; signs and significance of β_N and β_{N^2} are the cross-model summary. For ablations and the mechanism probes we also record the target’s rank, entropy, $\Sigma P(\text{synonyms})$ over a curated semantic field, and $\Sigma P(\text{unrelated pool})$. A separate deterministic classifier labels per-(model, frame, word) curves as *satiation-like*, *monotonic-priming*, *plateau*, or *mixed*; thresholds and the full regression are in Appendix B.

4 Probe Position Reverses Repetition Dynamics

4.1 Adjacent vs displaced: the dissociation

On the adjacent slot the three baseline models behave broadly as a priming account would predict. $P(\text{target})$ climbs with N and plateaus by $N \approx 20$ (Figure 2a), and the MLM never registers a single *satiation-like* word. Move the slot off the repeated block — same prefix, different probe position — and the curve no longer climbs. It rises through an early peak and falls back by $N=30$ (Figure 2b,c) in 12 of 13 panel models; per-word drops clear zero in all 13 (§4.2). Because the adjacent baseline shows that local continuation is still on the table, the displaced decline cannot be read as generic capacity loss; it has to be about the slot.

4.2 Panel-wide displaced result

F0 drops on the full panel range from 0.138 [0.114, 0.164] in SmoLLM2-1.7B to 0.733 [0.706, 0.761] in Qwen2.5-14B (Table 1). Every per-word drop has a bootstrap interval that clears zero. This is our headline result: the displaced inverted-U is a 13-of-13 effect under the per-word statistic we adopt as primary. As a secondary descriptive observation, 12 of 13 mean curves also turn over before $N=30$, with per-model peaks at $N \in \{1, 3, 5, 10, 20\}$; the exception is SmoLLM2-1.7B, whose per-word drops remain positive (0.138 [0.114, 0.164]) but whose population-mean curve continues to climb. We treat the per-word statistic as primary because it does not assume a single peak location across heterogeneous targets.

Frame heterogeneity. Across F0–F4 the per-word drop is reliably positive in all 65 of 65 (model, frame) cells. The 13-model mean drop is also

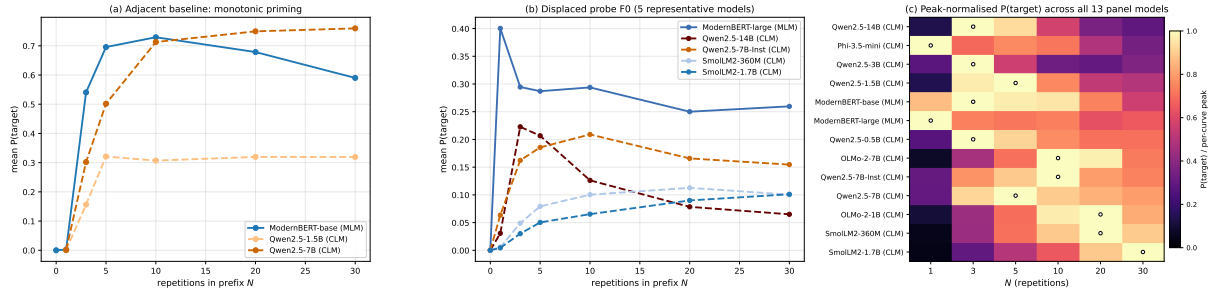


Figure 2: **Same prefix, different probe position.** (a) Adjacent baseline on three models: priming, plateau by $N \approx 20$. (b) Displaced probe (F0), five representative models spanning the panel’s peak- N range. Solid: MLM; dashed: CLM. (c) Peak-normalised $P(\text{target})$ across all 13 panel models, rows sorted by F0 drop magnitude; white dot marks each row’s peak N . Per-model numbers in Table 1.

tightly clustered (F0 0.460, F1 0.451, F2 0.483, F3 0.458, F4 0.434), so the headline result does not hinge on which frame we single out — F0 happens to be one of the larger frames, but none of them is the obvious winner.

4.3 Quadratic regression and frequency control

A cross-model OLS quadratic on shared-core cells gives the two non-monotonic predictions a panel-level test. The N coefficient is significantly positive ($\beta_N = +0.212$, $t = +36.8$) and the N^2 coefficient is significantly negative ($\beta_{N^2} = -0.00610$, $t = -40.2$); both survive clustering of SEs by target word, and the implied vertex sits at $N^* \approx 17.4$.¹ A targeted frequency-modulation regression finds a small but reliable Zipf effect on per-word drop ($\beta_{\text{Zipf}} = -0.036$, $t = -3.5$); the (model, frame) fixed effects nevertheless account for roughly $60\times$ more variance than Zipf does, so frequency is a modulator rather than the driver. Full coefficient tables and the cluster-robust panel regression are in Appendix B, Tables 3–4.

5 Exact Lexical Repetition Drives the Effect

5.1 Exact lexical repetition is the active variable

What in the prefix is doing the work? We swap the $N=30$ target block for five alternatives on the F0 displaced probe: a truncated $N=3$ repeat of the same target, no block at all, a length-matched random filler, a semantic-neighbour filler, and an unrelated-repeat block (`'table'` $\times 30$; constructions in Appendix D).

¹This is an average-trajectory vertex, not any individual model’s peak.

Four of the five swaps kill the effect outright. Random filler, deletion, and the unrelated-repeat block all push $P(\text{target})$ below 10^{-3} on all four ablation models. The interesting case is the semantic-neighbour filler. It is the only condition that *activates* the target’s semantic field: $\Sigma P(\text{synonyms})$ climbs to 0.316 in ModernBERT-large and to 0.10–0.16 in the three CLMs (Figure 3a,b). But $P(\text{target})$ itself stays at 0.012–0.031, well below the full-repeat range of 0.042–0.260. Semantic-field activation and target-specific priming come apart cleanly. Whatever the displaced inverted-U is, it is not driven by sequence length, not by repetition in general, and not by exposure to neighbours of the target — it is exact lexical repetition of the target itself, interacting with the displaced readout. That separation is what motivates the mechanism probes in §7.

5.2 Truncation as a within-thesis operationalisation

If the displaced curve really turns over before $N=30$, then truncating the prefix back to $N=3$ should land closer to the per-model peak and ought to raise $P(\text{target})$. Two of four ablation models confirm this directly: ModernBERT-large gains $+0.034$ [0.016, 0.056] with the target’s rank improving from 35 to 7, and Qwen2.5-1.5B gains $+0.059$ [0.046, 0.072] with rank $75 \rightarrow 7$. Qwen2.5-7B moves in the same direction in point estimate ($+0.015$, rank $10 \rightarrow 4$) but its CI $[-0.005, 0.035]$ just clips zero. OLMo-2-7B initially looked like a puzzle. Its truncation contrast is -0.015 $[-0.022, -0.009]$ — a real negative. But the same checkpoint shows a textbook peak-then-decay shape on the full $N=1 \rightarrow 30$ grid (per-word drop 0.426, Table 1); its peak just sits at $N=10$, so cutting to $N=3$ overshoots. The truncation prediction

Model	Obj.	Peak N	Peak P	$P(N=30)$	Drop	95 % CI of drop
Qwen2.5-14B (Qwen Team, 2025)	CLM	3	0.223	0.065	0.733	[0.706, 0.761]
Qwen2.5-1.5B (Qwen Team, 2025)	CLM	5	0.125	0.062	0.693	[0.654, 0.733]
Qwen2.5-7B-Instruct (Qwen Team, 2025)	CLM	10	0.209	0.155	0.549	[0.507, 0.592]
Phi-3.5-mini (Abdin et al., 2024)	CLM	1	0.001	0.000	0.523	[0.485, 0.558]
ModernBERT-base (Warner et al., 2025)	MLM	3	0.668	0.372	0.516	[0.481, 0.549]
Qwen2.5-7B (Qwen Team, 2025)	CLM	5	0.134	0.106	0.461	[0.422, 0.502]
Qwen2.5-3B (Qwen Team, 2025)	CLM	3	0.109	0.040	0.434	[0.397, 0.477]
OLMo-2-1B (OLMo Team, 2025)	CLM	20	0.039	0.033	0.434	[0.387, 0.472]
OLMo-2-7B (OLMo Team, 2025)	CLM	10	0.058	0.042	0.426	[0.385, 0.468]
ModernBERT-large (Warner et al., 2025)	MLM	1	0.400	0.260	0.423	[0.388, 0.462]
Qwen2.5-0.5B (Qwen Team, 2025)	CLM	3	0.187	0.132	0.329	[0.294, 0.363]
SmolLM2-360M (Ben Allal et al., 2025)	CLM	20	0.113	0.100	0.327	[0.291, 0.367]
SmolLM2-1.7B (Ben Allal et al., 2025)	CLM	30	0.101	0.101	0.138	[0.114, 0.164]

Table 1: Displaced-probe F0 summary across the open-access panel (per-model target sets of 182–258 single-token-valid words). *Peak N* , *Peak P* from the word-averaged mean curve; $P(N=30)$ the same curve at $N=30$. *Drop* is the per-word mean of $(\text{peak}_w - P(N=30)_w)/\text{peak}_w$; 95 % CI bootstrapped over words (500 resamples, seed 42). Every model has positive drop with CI excluding zero; SmolLM2-1.7B weakest, Qwen2.5-14B strongest.

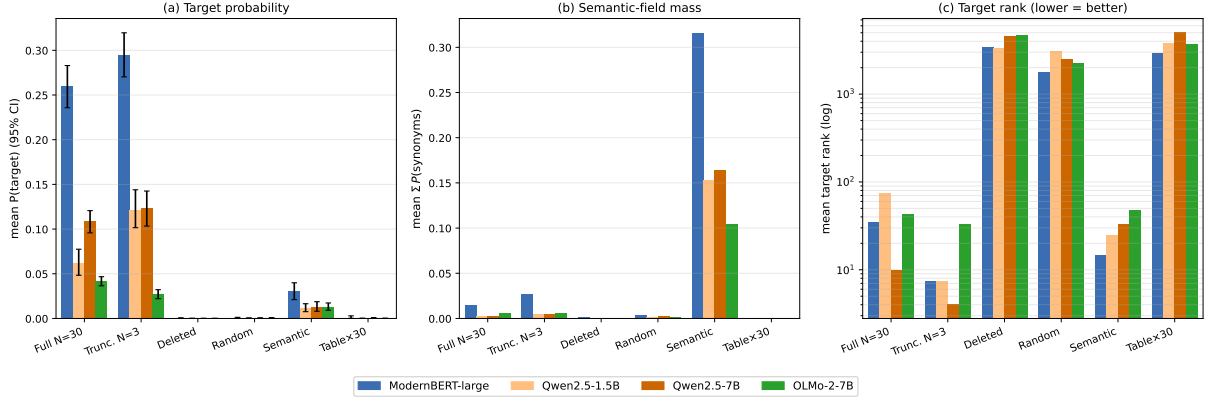


Figure 3: **Six-condition causal ablation, F0 displaced, four models** (shared legend below). (a) Mean $P(\text{target})$ per condition, 95% bootstrap CIs. (b) $\Sigma P(\text{synonyms})$ per condition. (c) Mean target rank (log scale) per condition. Non-semantic conditions use the per-model single-token-valid set; the semantic-neighbour filler and $\Sigma P(\text{synonyms})$ use targets with curated semantic fields. Exact means in Appendix D, Table 6.

Model	truncate – full	95 % CI	Mean target rank: full → truncate
ModernBERT-large	+0.034	[0.016, 0.056]	35 → 7
Qwen2.5-1.5B	+0.059	[0.046, 0.072]	75 → 7
Qwen2.5-7B	+0.015	[−0.005, 0.035]	10 → 4
OLMo-2-7B	−0.015	[−0.022, −0.009]	43 → 33

Table 2: Causal ablation: truncating $N=30$ to $N=3$ on F0 displaced. Contrast is the per-word mean $P(\text{target} | N=3) - P(\text{target} | N=30)$; 95 % CI bootstrapped over per-model single-token-valid targets. Two of four models replicate with CI excluding zero (ModernBERT-large $\approx 5 \times$ rank improvement, Qwen2.5-1.5B $\approx 11 \times$); Qwen2.5-7B’s point estimate is positive but CI includes zero. OLMo-2-7B’s CI excludes zero in the opposite direction: its displaced peak sits at $N=10$, so $N=3$ lands before the peak.

holds where the peak is in fact near $N=3$.

5.3 Peak-relative truncation contrast

The truncation contrast in §5.2 cuts the prefix from $N=30$ back to $N=3$ and asks whether $P(\text{target})$ rises. The prediction it tests is really about cutting the prefix back near the per-model peak, and $N=3$ is only a stand-in for that. Three of the four ablation

models have their displaced peak at or near $N=3$ (ModernBERT-large at $N=1$, Qwen2.5-1.5B and Qwen2.5-7B at $N=5$), which is why the contrast in Table 2 points in the expected direction for those three. OLMo-2-7B’s peak sits at $N=10$, $N=3$ sits well before it, and the same fixed- $N=3$ cut overshoots in the other direction.

Re-running the contrast at each model’s own peak gives a cleaner read of the same data. The per-word contrast $P(\text{target} \mid N=\text{peak}) - P(\text{target} \mid N=30)$ is $+0.141$ [$+0.114, +0.169$] (ModernBERT-large, $N=30 \rightarrow 1$), $+0.063$ [$+0.050, +0.076$] (Qwen2.5-1.5B, $N=30 \rightarrow 5$), $+0.028$ [$+0.011, +0.047$] (Qwen2.5-7B, $N=30 \rightarrow 5$), and $+0.016$ [$+0.010, +0.023$] (OLMo-2-7B, $N=30 \rightarrow 10$). All four are positive with bootstrap intervals clearing zero, including Qwen2.5-7B (whose $N=3$ contrast in Table 2 only narrowly clipped zero) and OLMo-2-7B (whose $N=3$ contrast was a real negative).

This is not a new experiment so much as a sharper read of the same per- N grid that produced Table 1. The truncation prediction the displaced inverted-U makes is “cut the prefix back near the peak and $P(\text{target})$ goes up.” It is not “ $N=3$ is privileged.” Under the peak-relative reading, OLMo-2-7B stops being an exception to the truncation claim — it was only ever an exception to the $N=3$ -specific operationalisation of it.

6 Boundaries and Heterogeneity

The displaced inverted-U is broad enough that we can ask where it is weakest, and what training-side knobs change its magnitude.

6.1 Within-family heterogeneity

The Qwen2.5 family lets us read scale at fixed tokenizer and training recipe. F0 drops are 0.329 (0.5B), 0.693 (1.5B), 0.434 (3B), 0.461 (7B), 0.733 (14B), and 0.549 (7B-Instruct). Bigger does not monotonically mean stronger or weaker: 1.5B and 14B sit at the top of the family, 0.5B and 7B-base near the bottom, and instruction tuning at 7B *increases* the drop from 0.461 to 0.549. The same direction holds in every one of the five frames: across F0–F4 the Qwen2.5-7B-Instruct drops are $\{0.549, 0.512, 0.494, 0.376, 0.281\}$ versus base $\{0.461, 0.282, 0.393, 0.283, 0.209\}$ (Table 5). Cross-family results land inside the Qwen envelope — Phi-3.5-mini 0.523; OLMo-2 0.434 at 1B and 0.426 at 7B; SmolLM2 0.327 at 360M and 0.138 at 1.7B — so the inverted-U appears at every scale we test but its magnitude is not a clean function of size. SmolLM2-1.7B is the only checkpoint whose mean displaced curve does not visibly invert through $N=30$, although its per-word drops are still positive (the headline statistic

holds). Its *satiation-like* share (≤ 0.18) sits well below the rest of the panel (0.4–0.96). The within-family SmolLM2 depth-vs-width comparison in Appendix C also does not give us a clean account of the SmolLM2-1.7B exception.

6.2 Cross-lingual scope check

A reasonable concern after the English panel is that the displaced inverted-U could be English-specific, or specific to WordPiece-like tokenisation, or specific to the particular English frames we use. The cleanest way to rule those out is to rerun the displaced probe unchanged in different languages and to do it on tokenisers that do not look like WordPiece. We probe four open-access multilingual checkpoints — XLM-R-base, XLM-R-large (Conneau et al., 2020), Qwen2.5-1.5B, and Qwen2.5-7B (Qwen Team, 2025) — on Spanish, Chinese, German, and French single-token noun lists, using three hand-translated frames per language and $n_w=28$ –49 per (model, language) cell. XLM-R’s BPE retains no Chinese single-token candidates after filtering, so XLM-R Chinese drops out for both base and large; what remains is fourteen (model, language) combinations and 42 (model, language, frame) cells in total.

Every one of those 42 cells shows a positive per-word drop with its bootstrap interval clearing zero, and 30 of 42 also turn the mean curve over before $N=30$. The twelve cells whose mean curve peaks at $N=30$ split cleanly into two groups: five are Qwen2.5 cells (1.5B on German F2, French F1, French F2; 7B on German F2 and French F2), and seven are XLM-R-large cells (all six of its non-German cells plus German F0). Their per-word drops are still positive. Magnitude in Spanish is large but spread, with F0 drops between 0.308 (XLM-R-large) and 0.853 (Qwen2.5-1.5B); German (0.354–0.814), French (0.443–0.823), and Chinese (0.435 Qwen2.5-7B / 0.481 Qwen2.5-1.5B) span similar ranges. We do not attempt a strong typological reading of the language-by-language ordering. Whether the magnitude differences reflect tokenizer choices, morphological richness, or particular frame translations is something this panel cannot tell us; what it does let us say is that the displaced inverted-U is not an artefact of any one of those three factors, since the shape recurs on two multilingual encoders (XLM-R-base and XLM-R-large) and two multilingual decoders (Qwen2.5-1.5B and Qwen2.5-7B), in four typologically different target languages, every cell we

measured. Per-(model, language, frame) numbers are in Appendix E, Table 7.

7 Internal Correlates

The behavioural picture is by now clear; the next question is what inside the model moves alongside it. We run two attention probes and a slot-state probe on four panel models chosen to span both objectives and three scales — ModernBERT-large, Qwen2.5-1.5B, Qwen2.5-7B, and OLMo-2-7B. These probes are correlational, not causal (Tenney et al., 2019; Hu and Levy, 2023). We record internal patterns that co-move with the behavioural effect. We do not claim these patterns cause the behavioural effect, and the probes below are not designed to distinguish among mechanisms that produce the same co-movement.

Total block attention grows with N . Per-target-token attention falls with N in all four mechanism models (Figure 4a), as budget conservation would predict if a fixed budget were divided over N keys. The total attention the model assigns to the repeated block, however, does not stay fixed. In three of four models the block budget grows with N (Figure 4b): Qwen2.5-1.5B from ≈ 0.11 at $N=3$ to ≈ 0.47 at $N=30$; Qwen2.5-7B from 0.06 to 0.29; OLMo-2-7B from 0.03 to 0.24. ModernBERT-large is the exception, with a roughly flat block budget (0.09 to 0.14; per- N table in Appendix F). The per-token fall in the three CLMs is therefore sub-dilution: per-token attention drops because the budget is divided across more tokens, but the budget itself grows.

The total-block-budget pattern is robust across layers (Figure 5). In the three CLM models, per-target-token attention sits above the per-layer $1/N$ reference at most depths, indicating sub-dilution is a stack-wide property rather than a final-layer artifact. ModernBERT-large is the exception, with super-dilution concentrated in its upper layers, consistent with its roughly flat block-budget profile in the last-layer analysis. The displaced inverted-U’s attention correlate dissociates by training objective in this subset: CLMs allocate more total attention to the repeated block as N grows; the MLM does not.

Slot-state target probability tracks the behavioural inverted-U. On the 23-word mechanism subset, the model’s logit-head probability for the target shows a clean inverted-U on the three CLMs: Qwen2.5-1.5B peaks at $N=3$ (0.138),

Qwen2.5-7B peaks around $N=5-10$ (0.18–0.20), and OLMo-2-7B peaks at $N=10$ (0.062); all decay through $N=30$ (Figure 4c). ModernBERT-large peaks earliest at $N=1$ (0.354) and decays unevenly.

Why might block attention grow with N ? Multiple internal mechanisms are compatible with this pattern — softmax cooperation between repeated keys, attention-sink dynamics (Xiao et al., 2024; Gu et al., 2025) under a displaced-query setup, or autoregressive-specific recency priors. Our measurements are correlational throughout and do not distinguish among these; the question of why CLMs allocate growing budgets to repeated blocks remains open.

8 Discussion

The methodological lesson sits in a single contrast. Hold the repeated-target prefix fixed, move the prediction slot off the block, and the curve no longer climbs: it peaks early and falls back. The same prefix produces priming on the adjacent slot in every baseline we test, so the displaced decline cannot be reduced to long-context capacity loss. This matters concretely for cloze probes of factual knowledge, semantic priming studies, in-context learning analyses that vary demonstration count, and any benchmark that scores predictions at a slot adjacent to the variation. Each implicitly treats slot position as orthogonal to how the variation affects prediction; the displaced inverted-U says the assumption is wrong, and the cross-lingual replication argues this isn’t an English- or tokenizer-specific concern.

The substantive claim is that repetition helps when the readout slot rides surface continuation, and hurts when the target has to be retrieved out of a fresh clause. The six-condition ablation places the active variable on exact lexical repetition, not prefix length and not semantic neighbourhood activation.

Where is the diagnostic weakest? SmolLM2-1.7B is the only checkpoint whose mean displaced curve does not visibly invert through $N=30$, though its per-word drops remain positive — the headline statistic survives across the full 13-model panel. The within-family comparison with SmolLM2-360M cuts against parameter count: the smaller, deeper model shows a clearer inversion than the larger, shallower one. The same direction shows up on BERT-tiny/mini/small (Appendix C): the 2-layer BERT-tiny shows no inverted-U, while 4-layer BERT-mini and BERT-small do (drops

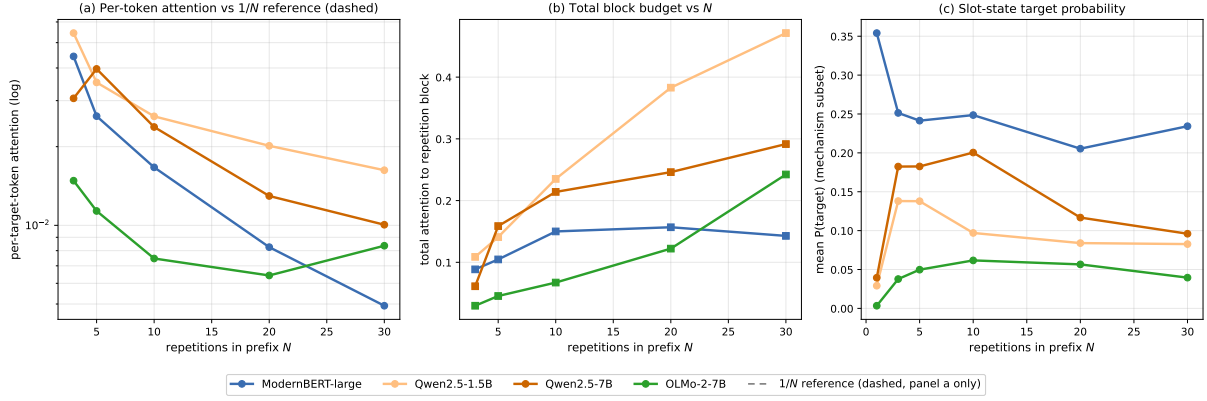


Figure 4: **Internal correlates on four panel models** (10/23-word mechanism subsets, F0 displaced; shared legend). (a) Per-target-token attention (solid, log) vs each model’s $1/N$ reference (dashed, anchored at smallest N): attention drops with N in every model. ModernBERT-large tracks its $1/N$ reference; the three CLMs sit above theirs (sub-dilution). (b) Total block attention: grows with N for the three CLMs (sub-dilution); plateaus for ModernBERT-large. (c) Slot-state target probability (23-word subset): in the three CLMs, slot-state target probability tracks the behavioural inverted-U; ModernBERT-large peaks earliest and decays unevenly.

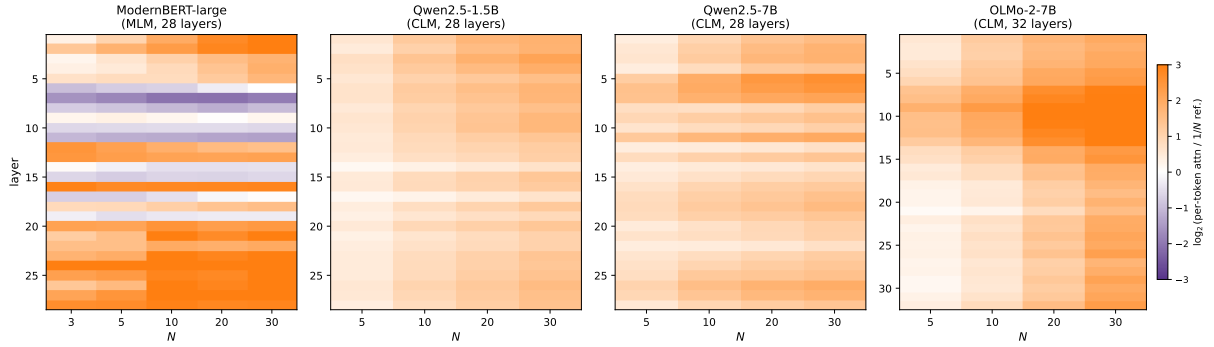


Figure 5: **Layer-wise per-target-token attention vs the per-layer $1/N$ reference**, displaced slot, four mechanism models on the 50-word stratified subset. Cells show $\log_2(\text{per-token attn} / 1/N \text{ ref. at smallest } N)$. Purple = super-dilution (per-token attention falls below the budget-dilution baseline); orange = sub-dilution (the two endpoints differ in luminance for grayscale readers). ModernBERT-large super-dilutes in its upper layers; Qwen2.5-1.5B, Qwen2.5-7B, and OLMo-2-7B sit predominantly above their $1/N$ references across the stack.

0.521 and 0.478). Depth, not parameter count, looks like the more relevant axis.

The internal correlate has a similar texture. Per-target-token attention drops with N in every model, which is what budget conservation predicts on its own. What it does not predict is the total attention the model assigns to the repeated block, which grows with N in the three CLM models and stays roughly flat in ModernBERT-large. The MLM/CLM split lines up with training objective rather than scale, and is consistent with autoregressive training doing something specific with repeated keys that an MLM does not. A single MLM is not enough to tell candidates apart, and the layer-targeted interventions that would distinguish them remain open.

As a downstream-style spot check, we piloted a

LAMA-style probe on the four mechanism models: the answer is stated upfront, an N -copy block of the target sits between, and the model is asked to re-recall (Appendix G). With this second retrieval route available, the displaced attenuation is muted (per-word drops 0.058–0.136 versus 0.42–0.69 in the main displaced probe), but three of four mechanism models still show positive drops with CI excluding zero (ModernBERT-large, Qwen2.5-1.5B, Qwen2.5-7B); OLMo-2-7B is the exception, climbing through $N=30$. The diagnostic carries detectable signal into the LAMA-style regime, attenuated by the explicit answer.

Whether the same shape transfers further, to free-form generation or multi-token targets, is outside the paper’s scope.

Limitations

Every target in our English panel passes a single-token filter in every tested tokenizer family, which excludes many naturalistic content words. The obvious extension — multi-token targets scored under probability-product or first-piece heuristics — is not a free generalisation: the headline shape statistic and the ablation conditions both assume a single readout position. The non-additive aggregation claim is bounded accordingly to single-token cloze and next-token prediction under the displaced-readout frame structure of §3; whether the same shape appears in free-form generation, multi-hop retrieval, or under multi-token scoring is a separate question.

The two probes are unevenly covered. The adjacent baseline in Figure 2a runs on three models (one MLM, one small CLM, one larger CLM), not on all 13; the dissociation requires the contrast to hold at all, not model-by-model, but extending the adjacent probe to the full panel would strengthen the argument. The cross-lingual frames F0–F2 are hand-translated to idiomatic equivalents, not calibrated for matched syntactic predictability, so frame-by-frame magnitudes across languages may partly reflect translation choices; the panel-level claim (42 of 42 cells positive) does not depend on those comparisons. SmolLM2-1.7B is an exception to the secondary descriptive observation (mean-curve inversion before $N=30$) but not to the headline statistic (positive per-word drop with CI excluding zero). We do not have a clean account of why this particular checkpoint’s mean curve fails to invert. Per-word drops are computed only on words with $\text{peak}_w > 0$; Phi-3.5-mini’s absolute $P(\text{target})$ is much lower than the rest of the panel ($\text{peak}_P \approx 0.001$ at the mean-curve level), and its 13-of-13 contribution rests on relative collapse across this low-probability regime rather than on a meaningful absolute decay.

The internal-correlate evidence is correlational. Per-target-token attention drops, the slot-state inverted-U turns over, and the layer-wise picture is structured rather than diffuse — all co-move with the behaviour, but co-movement is not intervention. None of our probes manipulates the attention pattern or the displaced slot directly; calibrated interventions that mask the repeated block’s attention or perturb the slot-state representation remain open.

References

- Marah Abdin, Sam Ade Jacobs, Ammar Ahmad Awan, Jyoti Aneja, Ahmed Awadallah, Hany Awadalla, and 1 others. 2024. [Phi-3 technical report: A highly capable language model locally on your phone](#). *Preprint*, arXiv:2404.14219.
- Rishabh Agarwal, Avi Singh, Lei M. Zhang, Bernd Bohnet, Luis Rosias, Stephanie Chan, Ankesh Anand, Zaheer Abbas, Azade Nova, John D. Co-Reyes, Eric Chu, Feryal Behbahani, Aleksandra Faust, and Hugo Larochelle. 2024. Many-shot in-context learning. *Advances in Neural Information Processing Systems (NeurIPS)*.
- Federico Barbero, Andrea Banino, Steven Kapturowski, Dharshan Kumaran, João G. M. Araújo, Alex Vitvitskyi, Razvan Pascanu, and Petar Veličković. 2024. Transformers need glasses! information oversquashing in language tasks. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Loubna Ben Allal, Anton Lozhkov, Elie Bakouch, Gabriel Martín Blázquez, Guilherme Penedo, Lewis Tunstall, Andrés Marafioti, Hynek Kydlíček, Agustín Piqueres Lajarín, Vaibhav Srivastav, Joshua Lochner, Caleb Fahlgren, Xuan-Son Nguyen, Clémentine Fourrier, Ben Burtenshaw, Hugo Larcher, Haojun Zhao, Cyril Zakka, Mathieu Morlon, and 3 others. 2025. [SmolLM2: When smol goes big – data-centric training of a small language model](#). *Preprint*, arXiv:2502.02737.
- Sheila R. Black. 2004. Reviewing the past and envisioning the future: An overview of issues in semantic satiation. *The American Journal of Psychology*, 117(1):53–74.
- Alexis Conneau, Kartikay Khandelwal, Naman Goyal, Vishrav Chaudhary, Guillaume Wenzek, Francisco Guzmán, Edouard Grave, Myle Ott, Luke Zettlemoyer, and Veselin Stoyanov. 2020. Unsupervised cross-lingual representation learning at scale. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 8440–8451.
- Allyson Ettinger. 2020. What BERT is not: Lessons from a new suite of psycholinguistic diagnostics for language models. *Transactions of the Association for Computational Linguistics (TACL)*, 8:34–48.
- Xiangming Gu, Tianyu Pang, Chao Du, Qian Liu, Fengzhuo Zhang, Cunxiao Du, Ye Wang, and Min Lin. 2025. When attention sink emerges in language models: An empirical view. In *International Conference on Learning Representations (ICLR)*.
- Ari Holtzman, Jan Buys, Li Du, Maxwell Forbes, and Yejin Choi. 2020. The curious case of neural text degeneration. In *International Conference on Learning Representations (ICLR)*.
- Jennifer Hu and Roger Levy. 2023. Prompting is not a substitute for probability measurements in large

689	language models. In <i>Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)</i> .	744
690		745
691		
692	Gautier Izacard and Edouard Grave. 2021. Leveraging	746
693	passage retrieval with generative models for open	747
694	domain question answering. In <i>Proceedings of the</i>	748
695	<i>16th Conference of the European Chapter of the As-</i>	749
696	<i>sociation for Computational Linguistics (EACL)</i> .	750
697		751
698	Vladimir Karpukhin, Barlas Oğuz, Sewon Min, Patrick	752
699	Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and	753
700	Wen-tau Yih. 2020. Dense passage retrieval for open-	754
701	domain question answering. In <i>Proceedings of the</i>	755
702	<i>2020 Conference on Empirical Methods in Natural</i>	
703	<i>Language Processing (EMNLP)</i> .	
704	Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio	757
705	Petroni, Vladimir Karpukhin, Naman Goyal, Hein-	758
706	rich Küttler, Mike Lewis, Wen-tau Yih, Tim Rock-	759
707	täschel, Sebastian Riedel, and Douwe Kiela. 2020.	760
708	Retrieval-augmented generation for knowledge-	761
709	intensive NLP tasks. In <i>Advances in Neural Informa-</i>	762
710	<i>tion Processing Systems (NeurIPS)</i> .	763
711		
712	Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. 2024. Lost in the middle: How language models use long contexts. <i>Transactions of the Association for Computational Linguistics (TACL)</i> , 12:157–173.	764
713		765
714		766
715	Yao Lu, Max Bartolo, Alastair Moore, Sebastian Riedel, and Pontus Stenertorp. 2022. Fantastically ordered prompts and where to find them: Overcoming few-shot prompt order sensitivity. In <i>Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL)</i> .	767
716		768
717		769
718		770
719		
720		
721	Sewon Min, Xinxu Lyu, Ari Holtzman, Mikel Artetxe, Mike Lewis, Hannaneh Hajishirzi, and Luke Zettlemoyer. 2022. Rethinking the role of demonstrations: What makes in-context learning work? In <i>Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP)</i> .	771
722		772
723		773
724		774
725		775
726		776
727		777
728	Kanishka Misra, Allyson Ettinger, and Julia Rayz. 2020. Exploring BERT’s sensitivity to lexical cues using tests from semantic priming. In <i>Findings of the Association for Computational Linguistics: EMNLP 2020</i> .	778
729		779
730		780
731		781
732	OLMo Team. 2025. <i>2 OLMo 2 furious</i> . Preprint, arXiv:2501.00656.	782
733		783
734		784
735	Fabio Petroni, Tim Rocktäschel, Sebastian Riedel, Patrick Lewis, Anton Bakhtin, Yuxiang Wu, and Alexander Miller. 2019. Language models as knowledge bases? In <i>Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP)</i> , pages 2463–2473.	785
736		786
737		787
738		788
739		789
740	Maura Pilotti, John S. Antrobus, and Martin Duff. 1997. Repetition and the perceptual difficulty of unfamiliar foreign words. <i>The American Journal of Psychology</i> , 110(4):521–541.	790
741		791
742		792
743		793
		794
		795
		796
		797
		798
		799
		800
	Qwen Team. 2025. <i>Qwen2.5 technical report</i> . Preprint, arXiv:2412.15115.	
	Melanie Sclar, Yejin Choi, Yulia Tsvetkov, and Alane Suhr. 2024. Quantifying language models’ sensitivity to spurious features in prompt design or: How i learned to start worrying about prompt formatting. In <i>International Conference on Learning Representations (ICLR)</i> .	
	Lyle C. Smith and Raymond M. Klein. 1990. Semantic satiation affects category membership decision time but not lexical priming. <i>Memory & Cognition</i> , 18(1):46–51.	
	Robyn Speer. 2022. <i>wordfreq: v3.0</i> . Python package.	
	Ian Tenney, Patrick Xia, Berlin Chen, Alex Wang, Adam Poliak, R. Thomas McCoy, Najoung Kim, Benjamin Van Durme, Samuel R. Bowman, Dipanjan Das, and Ellie Pavlick. 2019. What do you learn from context? Probing for sentence structure in contextualized word representations. In <i>International Conference on Learning Representations (ICLR)</i> .	
	Xing Tian and David E. Huber. 2010. Testing an associative account of semantic satiation. <i>Cognitive Psychology</i> , 60(4):267–290.	
	Iulia Turc, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. Well-read students learn better: On the importance of pre-training compact models. <i>arXiv preprint arXiv:1908.08962</i> .	
	Benjamin Warner, Antoine Chaffin, Benjamin Clavié, Orion Weller, Oskar Hallström, Said Taghadouini, Alexis Gallagher, Raja Biswas, Faisal Ladhak, Tom Aarsen, Griffin Thomas Adams, Jeremy Howard, and Iacopo Poli. 2025. Smarter, better, faster, longer: A modern bidirectional encoder for fast, memory efficient, and long context finetuning and inference. In <i>Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)</i> , pages 2526–2547, Vienna, Austria. Association for Computational Linguistics.	
	Sean Welleck, Ilia Kulikov, Stephen Roller, Emily Dinan, Kyunghyun Cho, and Jason Weston. 2020. Neural text generation with unlikelihood training. In <i>International Conference on Learning Representations (ICLR)</i> .	
	Guangxuan Xiao, Yuandong Tian, Beidi Chen, Song Han, and Mike Lewis. 2024. Efficient streaming language models with attention sinks. In <i>International Conference on Learning Representations (ICLR)</i> .	
	Jin Xu, Xiaojiang Liu, Jianhao Yan, Deng Cai, Huayang Li, and Jian Li. 2022. Learning to break the loop: Analyzing and mitigating repetitions for neural text generation. In <i>Advances in Neural Information Processing Systems (NeurIPS)</i> .	
	Itay Yona, Ilia Shumailov, Jamie Hayes, and Yossi Gandelsman. 2025. Interpreting the repeated token phenomenon in large language models. In <i>Proceedings of the 42nd International Conference on Machine Learning (ICML)</i> .	

A Word List, Tokenizer Filtering, and Templates

Construction. We hand-curate 261 candidate English nouns and noun-like concepts. Each candidate is annotated with its Zipf log-frequency (Speer, 2022) and a semantic class. The list is balanced across four categorical cells (cross of concrete vs abstract reference, and high vs low frequency, with the high/low boundary at Zipf 5.0) and across twelve semantic classes.

Per-tokenizer filtering. Tokenizers fragment many candidates differently, so we construct a per-family target set by retaining candidates that encode to exactly one piece under that family’s tokenizer. For each candidate we try both the leading-space form (BPE / SentencePiece convention) and the no-space form (BERT WordPiece convention) and accept the candidate if either yields a single token. This produces target sets of 182 (Phi-3.5-mini) to 258 (SmolLM2) words. The shared core (the intersection across every panel tokenizer) is ≈ 168 words and is the basis for the cross-model OLS regression of §4.3.

Frame templates. *Adjacent probes* (no displaced frame):

- MLM: `<target × N> [MASK] .`
- CLM: `<target × N> (next-token slot).`

Displaced probes:

- MLM: `<target × N> . <frame> [MASK] .`
- CLM: `<target × N> . <frame> (next-token slot).`

Frames F0–F4 (English):

- F0: *The word I keep thinking about is ...*
- F1: *The topic was ...*
- F2: *I want to mention ...*
- F3: *Let me tell you about ...*
- F4: *The most important thing is ...*

Ablation block templates (replace `<target × 30>`). The full $N=30$ repeat (code: `full_repeat_N30`) keeps the original block; the truncated $N=3$ repeat (code: `truncate_to_N3`) keeps only the first three target copies; the deleted block (code:

`delete_repeat_block`) removes the block entirely; the length-matched random filler (code: `random_filler_same_length`) substitutes 30 random unrelated nouns drawn from a fixed neutral pool; the semantic-neighbour filler (code: `unique_semantic_filler`) substitutes 30 semantic-neighbour tokens sampled from the target’s curated semantic field, excluding the exact target; and the unrelated-repeat block (code: `repeat_unrelated_N30`) substitutes 30 copies of the fixed token ‘table’.

B Statistical Analysis: Full Tables

Quadratic regression — coefficients. The headline OLS quadratic on shared-core $N \geq 1$ cells ($n = 70,590$) yields the parameter table below; cluster-robust SEs (clustered by target word). Reference levels: model ModernBERT-base, frame F0, category abstract-high. $R^2 = 0.550$.

Term	Coefficient
Intercept	−0.725
N	+0.2118
N^2	−0.00610
Zipf log-frequency	−0.191
model: ModernBERT-large	−0.43
model: Qwen2.5-0.5B	−3.10
model: Qwen2.5-1.5B	−3.51
model: Qwen2.5-3B	−2.84
model: Qwen2.5-7B	−1.95
model: Qwen2.5-7B-Instruct	−2.10
model: Qwen2.5-14B	−2.05
model: SmolLM2-360M	−3.45
model: SmolLM2-1.7B	−2.82
model: Phi-3.5-mini	−7.34
model: OLMo-2-1B	−3.63
model: OLMo-2-7B	−2.90
frame: F1	−1.77
frame: F2	−2.29
frame: F3	−1.12
frame: F4	−2.27
category: abstract-low	−0.63
category: concrete-high	−0.27
category: concrete-low	−0.84

Table 3: OLS quadratic regression coefficients on $\log P(\text{target})$ over the shared-core panel, rounded for display. Both N and N^2 signs match the inverted-U prediction; the largest fixed effects are by model.

Pattern classification thresholds. Each per-word N -curve is classified by deterministic rules. Let $\text{peak}_w = \max_{N \in \{1,3,5,10,20,30\}} P(\text{target} \mid w, N)$ and let N_w^* denote the argmax. A (model, frame, word) cell is classified as

- *satiation-like* if $N_w^* \in \{1, 3, 5, 10\}$ and $(\text{peak}_w - P(N=30 | w)) / \text{peak}_w > 0.30$;
- *monotonic-priming* if the curve is non-decreasing in N and $P(N=30 | w) > 1.05 \cdot P(N=1 | w)$;
- *plateau* if the curve is approximately monotonic with $P(N=30 | w) \geq 0.7 \cdot \text{peak}_w$;
- *mixed* otherwise.

Across the 16,230 (model, frame, word) cells, 8,964 (55%) are *satiation-like*, 5,129 (32%) are *plateau*, 1,089 (7%) are *monotonic-priming*, and 1,048 (6%) are *mixed*. Per-(model, frame) shares appear in the rightmost column of Table 5.

Frequency confound regression. Table 4 reports the targeted regression $\text{drop}_w \sim \text{Zipf} + \text{category} + \text{model} + \text{frame}$ on $n=16,230$ word-frame cells with cluster-robust SEs by word. The full regression has $R^2=0.191$; Zipf alone has $R^2=0.003$ and Zipf+category together $R^2=0.004$, while model+frame fixed effects alone account for $R^2=0.186$, so the fixed effects explain $\approx 60\times$ more variance than Zipf alone. The Zipf coefficient is small but reliably negative under clustering (-0.036 , $t = -3.5$, $p < 10^{-3}$), so high-frequency words decay slightly less than low-frequency ones; the category coefficients lose individual significance under clustering (concrete-low: $t = -1.56$, $p = 0.12$; others $|t| < 1.5$), so we do not interpret per-category differences in decay magnitude. **Frequency modulates the magnitude of the decay but does not explain it away.**

Term	Coefficient
Zipf log-frequency	-0.036
category: abstract-low	+0.015
category: concrete-high	-0.012
category: concrete-low	-0.029
R^2 (full, with model+frame):	0.191
R^2 (Zipf only):	0.003
R^2 (Zipf + category only):	0.004
R^2 (model + frame only):	0.186
$n = 16,230$ word-frame cells	

Table 4: Frequency-confound regression $\text{drop}_w \sim \text{Zipf} + \text{category} + \text{model} + \text{frame}$; cluster-robust SEs (clustered by target word). Zipf has the right-sign coefficient but explains an order of magnitude less variance than the (model, frame) fixed effects.

Per-(model, frame) summary — complete table. Table 5 reports peak N , peak P , $P(N=30)$,

drop, bootstrap CI, and *satiation-like* share for every (model, frame) cell (65 rows: 13 models $\times$ 5 frames). SmolLM2-1.7B is the only row block whose mean curve does not invert on any frame; its per-word drops remain positive but the *satiation-like* share never exceeds 0.18.

C Within-Family Architecture Contrast

The two SmolLM2 checkpoints provide a small natural depth-vs-capacity dissociation. SmolLM2-360M has 32 transformer blocks at hidden size 960 (15 attention heads, 2560 intermediate dimension) and an F0 drop of 0.327. SmolLM2-1.7B has fewer blocks (24) at a wider hidden size (2048, 32 attention heads, 8192 intermediate) and an F0 drop of 0.138. The larger checkpoint has more parameters but is shallower, and shows the weaker effect; the smaller, deeper checkpoint shows the stronger effect. Within the SmolLM2 family, depth therefore correlates with displaced-inverted-U magnitude in the opposite direction from raw parameter count. We treat this as suggestive rather than conclusive (a single family with one depth-vs-width contrast is not a controlled scaling law).

BERT-tiny / mini / small. The same direction shows up on the older Turc et al. (2019) BERT family, which we ran on the F0 displaced probe as an auxiliary check. BERT-tiny (2 layers, hidden 128, 4.4M params) shows no inverted-U: F0 peak $N=30$, per-word drop 0.001 [0.000, 0.003]. BERT-mini (4 layers, hidden 256, 11M) shows a clear inverted-U: F0 peak $N=3$, per-word drop 0.521 [0.471, 0.565]. BERT-small (4 layers, hidden 512, 29M) also shows a per-word inversion with CI clearing zero (0.478 [0.426, 0.528]), though its mean curve peaks at $N=30$. The transition between absent and present sits at the depth jump from 2 to 4 layers; doubling hidden size at fixed depth (mini $\rightarrow$ small) leaves the per-word drop essentially unchanged. This is consistent with the SmolLM2 within-family direction, but neither comparison is a controlled scaling law on its own.

D Causal Ablation Construction

Condition definitions. On the F0 displaced probe, each input replaces the 30-target-repeat block with one of six conditions. The non-semantic conditions are run over the per-model single-token-valid set; the semantic-neighbour filler and the semantic-field mass are restricted to targets with curated semantic dictionaries. The

Model	Frame	Peak N	Peak P	$P(N=30)$	Drop	95% CI of drop	Sat. share
ModernBERT-base	F0	3	0.668	0.372	0.516	[0.481, 0.549]	0.74
	F1	5	0.292	0.107	0.728	[0.696, 0.761]	0.90
	F2	3	0.132	0.044	0.696	[0.665, 0.728]	0.91
	F3	3	0.329	0.091	0.787	[0.759, 0.812]	0.93
	F4	3	0.119	0.034	0.832	[0.805, 0.855]	0.96
ModernBERT-large	F0	1	0.400	0.260	0.423	[0.388, 0.462]	0.65
	F1	10	0.210	0.141	0.455	[0.415, 0.495]	0.56
	F2	10	0.062	0.050	0.382	[0.348, 0.422]	0.46
	F3	10	0.146	0.086	0.551	[0.514, 0.585]	0.74
	F4	1	0.160	0.084	0.570	[0.534, 0.604]	0.76
Qwen2.5-0.5B	F0	3	0.187	0.132	0.329	[0.294, 0.363]	0.53
	F1	3	0.012	0.008	0.398	[0.350, 0.444]	0.51
	F2	3	0.014	0.006	0.589	[0.553, 0.621]	0.80
	F3	3	0.015	0.014	0.256	[0.218, 0.292]	0.36
	F4	30	0.005	0.005	0.247	[0.214, 0.280]	0.33
Qwen2.5-1.5B	F0	5	0.125	0.062	0.693	[0.654, 0.733]	0.81
	F1	5	0.023	0.014	0.684	[0.639, 0.730]	0.71
	F2	3	0.010	0.002	0.825	[0.797, 0.856]	0.92
	F3	3	0.039	0.008	0.856	[0.829, 0.884]	0.94
	F4	5	0.008	0.004	0.629	[0.585, 0.672]	0.72
Qwen2.5-3B	F0	3	0.109	0.040	0.434	[0.397, 0.477]	0.60
	F1	30	0.014	0.014	0.418	[0.372, 0.462]	0.52
	F2	3	0.006	0.004	0.481	[0.444, 0.516]	0.59
	F3	3	0.054	0.026	0.551	[0.515, 0.587]	0.71
	F4	3	0.008	0.003	0.436	[0.401, 0.471]	0.60
Qwen2.5-7B	F0	5	0.134	0.106	0.461	[0.422, 0.502]	0.52
	F1	30	0.029	0.029	0.282	[0.247, 0.320]	0.33
	F2	30	0.018	0.018	0.393	[0.354, 0.434]	0.51
	F3	30	0.045	0.045	0.283	[0.250, 0.321]	0.37
	F4	30	0.115	0.115	0.209	[0.171, 0.249]	0.18
Qwen2.5-7B-Instruct	F0	10	0.209	0.155	0.549	[0.507, 0.592]	0.55
	F1	5	0.027	0.024	0.512	[0.465, 0.553]	0.58
	F2	10	0.014	0.011	0.494	[0.452, 0.535]	0.58
	F3	30	0.088	0.088	0.376	[0.343, 0.419]	0.44
	F4	30	0.097	0.097	0.281	[0.236, 0.327]	0.29
Qwen2.5-14B	F0	3	0.223	0.065	0.733	[0.706, 0.761]	0.93
	F1	5	0.028	0.015	0.499	[0.464, 0.530]	0.71
	F2	5	0.018	0.010	0.479	[0.447, 0.512]	0.69
	F3	3	0.083	0.031	0.674	[0.645, 0.704]	0.89
	F4	5	0.021	0.006	0.545	[0.511, 0.577]	0.78
SmolLM2-360M	F0	20	0.113	0.100	0.327	[0.291, 0.367]	0.29
	F1	20	0.006	0.006	0.267	[0.233, 0.301]	0.27
	F2	20	0.009	0.008	0.365	[0.326, 0.407]	0.42
	F3	20	0.032	0.030	0.262	[0.223, 0.300]	0.28
	F4	20	0.003	0.003	0.328	[0.294, 0.365]	0.41
SmolLM2-1.7B	F0	30	0.101	0.101	0.138	[0.114, 0.164]	0.05
	F1	30	0.032	0.032	0.255	[0.221, 0.290]	0.14
	F2	30	0.019	0.019	0.206	[0.171, 0.240]	0.18
	F3	30	0.040	0.040	0.191	[0.161, 0.224]	0.09
	F4	30	0.026	0.026	0.213	[0.181, 0.247]	0.05
Phi-3.5-mini	F0	1	0.001	0.000	0.523	[0.485, 0.558]	0.75
	F1	1	0.000	0.000	0.459	[0.421, 0.497]	0.69
	F2	1	0.000	0.000	0.608	[0.574, 0.642]	0.92
	F3	10	0.000	0.000	0.333	[0.295, 0.373]	0.50
	F4	1	0.000	0.000	0.564	[0.528, 0.602]	0.84
OLMo-2-1B	F0	20	0.039	0.033	0.434	[0.387, 0.472]	0.48
	F1	20	0.013	0.013	0.500	[0.451, 0.545]	0.55
	F2	30	0.005	0.005	0.291	[0.244, 0.335]	0.29
	F3	20	0.024	0.022	0.301	[0.261, 0.343]	0.31
	F4	30	0.010	0.010	0.225	[0.187, 0.266]	0.24
OLMo-2-7B	F0	10	0.058	0.042	0.426	[0.385, 0.468]	0.43
	F1	10	0.022	0.015	0.412	[0.372, 0.454]	0.50
	F2	10	0.008	0.005	0.474	[0.433, 0.515]	0.56
	F3	10	0.041	0.022	0.537	[0.495, 0.579]	0.67
	F4	10	0.011	0.005	0.565	[0.529, 0.601]	0.72

Table 5: Per-(model, frame) summary (n_{words} ranges from 182 to 258). *Peak N* and *Peak P* are taken from the word-averaged mean curve. *Drop* is the per-word mean of $(\text{peak}_w - P(N=30)_w)/\text{peak}_w$; the 95% CI is bootstrapped over the per-model single-token-valid target words. *Sat. share* is the fraction of (word) curves classified as *satiation-like*. Model name shown on the F0 row of each block; blank below.

full $N=30$ repeat (code: `full_repeat_N30`) keeps thirty exact copies of the target. The truncated $N=3$ repeat (code: `truncate_to_N3`) keeps only the first three target copies. The deleted block (code: `delete_repeat_block`) is empty. The length-matched random filler (code: `random_filler_same_length`) is thirty random nouns sampled from a fixed neutral pool. The semantic-neighbour filler (code: `unique_semantic_filler`) is thirty semantic neighbours sampled from the target’s curated semantic field, excluding the exact target. The unrelated-repeat block (code: `repeat_unrelated_N30`) is thirty copies of ‘table’.

Per-(model, condition) summary. Table 6 reports means over the per-model target set for each condition. The full $N=30$ repeat and the truncated $N=3$ repeat are the only conditions with non-trivial $P(\text{target})$; the semantic-neighbour filler is the only condition that raises $\Sigma P(\text{synonyms})$ substantially, illustrating the separation between target-specific lexical priming and semantic-neighbourhood activation.

E Cross-Lingual Panel

Setup. Four open-access multilingual models are evaluated: FacebookAI/xlm-roberta-base (MLM, 279M) and FacebookAI/xlm-roberta-large (MLM, 561M) (Conneau et al., 2020); Qwen/Qwen2.5-1.5B and Qwen/Qwen2.5-7B (Qwen Team, 2025) (CLMs). Languages are Spanish, Chinese, German, and French; word lists are hand-curated single-token nouns of length ≥ 3 , balanced for concreteness/frequency where possible. Each candidate is filtered against the model’s own tokenizer using the same leading-space-first rule as the English panel, leaving 28–49 valid words per (model, language) cell. Frames F0–F2 are hand-translated to idiomatic equivalents of the English F0–F2 templates; F3/F4 are not translated.

Per-(model, language, frame) results. Table 7 reports peak N , peak P , drop, and 95% bootstrap CI for each cell. Mean-curve peaks land at $N=1$ for XLM-R-base, $N=3$ –10 for both Qwen2.5 sizes, and split between $N=1$ and $N=30$ for XLM-R-large. The twelve cells whose mean curve peaks

at $N=30$ are five Qwen2.5 cells (1.5B on German F2, French F1, French F2; 7B on German F2 and French F2) and seven XLM-R-large cells (all six non-German cells plus German F0); even there the per-word drop CI excludes zero.

Panel scope. The reported multilingual panel pairs the two XLM-R sizes with the two Qwen2.5 sizes; the four-model panel spans both objectives, two tokenizer families (XLM-R SentencePiece and Qwen BPE), and four languages.

F Mechanism Probes: Per- N and Layer-wise

Per- N attention table. Table 8 reports per-target-token attention at the displaced slot in the last layer (averaged across heads and target words), the $1/N$ reference anchored at the smallest evaluated N , and the total attention to the repetition block. In three of four mechanism models the total block budget grows monotonically with N ; ModernBERT-large is the exception, with a roughly flat block budget.

G LAMA-Style Pilot

Setup. Three answer-prompt templates of the form `<prefix> the answer is <target> . <block> . <suffix>` the answer is (A0–A2 paraphrase the prefix and suffix) are evaluated on the four mechanism models. The middle `<block>` is $N \in \{1, 3, 5, 10, 20, 30\}$ tokens long, drawn from one of four conditions: `target_repeat` (the target itself), `random_repeat` (random neutral nouns), `unrelated_repeat` (‘table’), or `semantic_repeat` (semantic neighbours). Each (model, template, condition, N) cell is run on a 50-word stratified subset of that model’s single-token-valid targets; per-word probabilities are averaged across the three templates.

Target-repeat curves. Table 9 reports the mean $P(\text{target})$ at each N for the `target_repeat` block (the directly comparable condition), and the per-word drop $(\text{peak}_w - P(N=30)_w)/\text{peak}_w$ with 95% bootstrap CI (500 resamples, seed 42). Three of four models replicate the displaced attenuation with CI excluding zero; the magnitude is smaller than the headline displaced effect because the pre-stated answer provides a second retrieval route.

Control conditions. The `unrelated_repeat` and `random_repeat`

Model	Condition	$P(\text{target})$	$\Sigma P(\text{syn.})$
ModernBERT-large	Full $N=30$ repeat	0.260	0.015
ModernBERT-large	Truncated $N=3$ repeat	0.294	0.027
ModernBERT-large	Deleted block	0.001	0.002
ModernBERT-large	Random filler	0.001	0.004
ModernBERT-large	Semantic-neighbour filler	0.031	0.316
ModernBERT-large	Unrelated-repeat block	0.001	0.001
Qwen2.5-1.5B	Full $N=30$ repeat	0.062	0.003
Qwen2.5-1.5B	Truncated $N=3$ repeat	0.121	0.005
Qwen2.5-1.5B	Deleted block	0.000	0.001
Qwen2.5-1.5B	Random filler	0.000	0.002
Qwen2.5-1.5B	Semantic-neighbour filler	0.012	0.154
Qwen2.5-1.5B	Unrelated-repeat block	0.000	0.001
Qwen2.5-7B	Full $N=30$ repeat	0.109	0.003
Qwen2.5-7B	Truncated $N=3$ repeat	0.124	0.005
Qwen2.5-7B	Deleted block	0.000	0.000
Qwen2.5-7B	Random filler	0.001	0.003
Qwen2.5-7B	Semantic-neighbour filler	0.013	0.164
Qwen2.5-7B	Unrelated-repeat block	0.000	0.000
OLMo-2-7B	Full $N=30$ repeat	0.042	0.006
OLMo-2-7B	Truncated $N=3$ repeat	0.027	0.006
OLMo-2-7B	Deleted block	0.000	0.000
OLMo-2-7B	Random filler	0.001	0.002
OLMo-2-7B	Semantic-neighbour filler	0.013	0.105
OLMo-2-7B	Unrelated-repeat block	0.000	0.000

Table 6: Mean $P(\text{target})$ and mean $\Sigma P(\text{synonyms})$ per (model, ablation condition) on the F0 displaced probe. Non-semantic conditions use the per-model single-token-valid set; the semantic-neighbour filler and the semantic-field mass are computed on the targets with curated semantic dictionaries. The semantic-neighbour filler is the only condition that raises the semantic-field mass substantially.

blocks (which break the target match) produce baseline $P(\text{target})$ in the 0.13–0.45 range across models without the inverted-U signature, confirming that the `target_repeat` effect is target-specific rather than a generic position artefact. The `semantic_repeat` block falls between these and the `target-repeat` block. Full per-condition curves are in the released data.

H Reproducibility

Pipeline. The full analysis pipeline is invoked through a single notebook that runs target-set construction, displaced evaluations, ablations, mechanism probes, figure rendering, and verification. The per-stage script-to-output mapping is documented in the reproducibility README. The mechanism and ablation probes for the four high-cost models are scoped to finish within ≈ 12 hours on a single A100 80GB.

Hardware. The full panel runs end-to-end on a single A100 80GB GPU in `bfloat16`. Smaller models (under 4B) finish in under ten minutes each; Qwen2.5-7B and the OLMo-2-7B mechanism battery take roughly 30–60 minutes; Qwen2.5-14B (displaced grid only) takes roughly ninety minutes.

Model checkpoints. All models are loaded from public HuggingFace checkpoints with no access tokens required. The exact checkpoint identifiers are `answerdotai/ModernBERT-base`, `answerdotai/ModernBERT-large`, `Qwen/Qwen2.5-0.5B`, `Qwen/Qwen2.5-1.5B`, `Qwen/Qwen2.5-3B`, `Qwen/Qwen2.5-7B`, `Qwen/Qwen2.5-7B-Instruct`, `Qwen/Qwen2.5-14B`, `HuggingFaceTB/SmolLM2-360M`, `HuggingFaceTB/SmolLM2-1.7B`, `microsoft/Phi-3.5-mini-instruct`, `allenai/OLMo-2-0425-1B`, and `allenai/OLMo-2-1124-7B`.

Random seeds. All bootstrap resampling uses seed 42 (500 resamples). All word-list stratifications and ablation samplings use seed 0. Model forward passes are deterministic.

Pinned dependencies. Exact versions of PyTorch, HuggingFace Transformers (≥ 4.48), pandas, numpy, scipy, statsmodels, matplotlib, and wordfreq are pinned at the snapshot used to produce the reference outputs.

Model	Lang.	Frame	n_w	Peak N	Peak P	Drop	95 % CI of drop
xlm-roberta-base	German	F0	46	1	0.784	0.515	[0.412, 0.618]
xlm-roberta-base	German	F1	46	1	0.302	0.561	[0.465, 0.662]
xlm-roberta-base	German	F2	46	20	0.006	0.502	[0.410, 0.583]
xlm-roberta-base	Spanish	F0	49	1	0.492	0.541	[0.431, 0.636]
xlm-roberta-base	Spanish	F1	49	1	0.210	0.737	[0.665, 0.803]
xlm-roberta-base	Spanish	F2	49	20	0.062	0.539	[0.442, 0.630]
xlm-roberta-base	French	F0	40	1	0.442	0.823	[0.731, 0.903]
xlm-roberta-base	French	F1	40	10	0.063	0.573	[0.447, 0.686]
xlm-roberta-base	French	F2	40	1	0.088	0.543	[0.432, 0.651]
xlm-roberta-large	German	F0	46	30	0.265	0.354	[0.257, 0.455]
xlm-roberta-large	German	F1	46	1	0.037	0.561	[0.455, 0.671]
xlm-roberta-large	German	F2	46	1	0.009	0.869	[0.793, 0.929]
xlm-roberta-large	Spanish	F0	49	30	0.207	0.308	[0.219, 0.389]
xlm-roberta-large	Spanish	F1	49	30	0.045	0.372	[0.273, 0.479]
xlm-roberta-large	Spanish	F2	49	30	0.002	0.749	[0.656, 0.830]
xlm-roberta-large	French	F0	40	30	0.208	0.443	[0.335, 0.555]
xlm-roberta-large	French	F1	40	30	0.012	0.115	[0.062, 0.181]
xlm-roberta-large	French	F2	40	30	0.004	0.639	[0.508, 0.756]
Qwen2.5-1.5B	German	F0	28	5	0.192	0.814	[0.718, 0.893]
Qwen2.5-1.5B	German	F1	28	5	0.036	0.624	[0.535, 0.714]
Qwen2.5-1.5B	German	F2	28	30	0.142	0.112	[0.034, 0.214]
Qwen2.5-1.5B	Spanish	F0	34	5	0.199	0.853	[0.785, 0.909]
Qwen2.5-1.5B	Spanish	F1	34	5	0.003	0.579	[0.479, 0.669]
Qwen2.5-1.5B	Spanish	F2	34	5	0.001	0.684	[0.600, 0.761]
Qwen2.5-1.5B	French	F0	29	3	0.105	0.777	[0.686, 0.861]
Qwen2.5-1.5B	French	F1	29	30	0.002	0.481	[0.361, 0.608]
Qwen2.5-1.5B	French	F2	29	30	0.001	0.568	[0.464, 0.680]
Qwen2.5-1.5B	Chinese	F0	49	3	0.007	0.481	[0.411, 0.549]
Qwen2.5-1.5B	Chinese	F1	49	3	0.007	0.367	[0.287, 0.451]
Qwen2.5-1.5B	Chinese	F2	49	10	0.007	0.384	[0.298, 0.472]
Qwen2.5-7B	German	F0	28	10	0.229	0.479	[0.366, 0.583]
Qwen2.5-7B	German	F1	28	5	0.040	0.610	[0.527, 0.706]
Qwen2.5-7B	German	F2	28	30	0.087	0.117	[0.047, 0.203]
Qwen2.5-7B	Spanish	F0	34	5	0.192	0.732	[0.646, 0.808]
Qwen2.5-7B	Spanish	F1	34	5	0.010	0.448	[0.374, 0.527]
Qwen2.5-7B	Spanish	F2	34	20	0.001	0.305	[0.222, 0.400]
Qwen2.5-7B	French	F0	29	5	0.152	0.486	[0.392, 0.587]
Qwen2.5-7B	French	F1	29	5	0.007	0.378	[0.291, 0.452]
Qwen2.5-7B	French	F2	29	30	0.002	0.202	[0.112, 0.306]
Qwen2.5-7B	Chinese	F0	49	5	0.027	0.435	[0.386, 0.483]
Qwen2.5-7B	Chinese	F1	49	3	0.012	0.575	[0.504, 0.639]
Qwen2.5-7B	Chinese	F2	49	5	0.027	0.387	[0.321, 0.455]

Table 7: Cross-lingual results. Four open-access multilingual models (XLM-R-base, XLM-R-large, Qwen2.5-1.5B, Qwen2.5-7B) on Spanish, Chinese, German, and French single-token noun lists, with three hand-translated frames per language. *Peak N* and *Peak P* are taken from the word-averaged mean curve; *Drop* is the per-word mean of $(\text{peak}_w - P(N=30)_w) / \text{peak}_w$, with bootstrap 95 % CI over the per-(model, language) word population. XLM-R Chinese is excluded for both base and large because XLM-R’s BPE leaves no Chinese single-token candidates. All 42 evaluated cells show a positive drop with CI excluding zero; 30 of 42 also have mean-curve peak before $N=30$.

Model	N	Per-token	$1/N$ ref.	Total to block
Qwen2.5-1.5B	3	0.054	0.054	0.109
Qwen2.5-1.5B	5	0.035	0.033	0.141
Qwen2.5-1.5B	10	0.026	0.016	0.235
Qwen2.5-1.5B	20	0.020	0.008	0.383
Qwen2.5-1.5B	30	0.016	0.005	0.471
Qwen2.5-7B	3	0.031	0.031	0.061
Qwen2.5-7B	5	0.040	0.018	0.159
Qwen2.5-7B	10	0.024	0.009	0.214
Qwen2.5-7B	20	0.013	0.005	0.246
Qwen2.5-7B	30	0.010	0.003	0.292
OLMo-2-7B	3	0.015	0.015	0.030
OLMo-2-7B	5	0.011	0.009	0.045
OLMo-2-7B	10	0.007	0.004	0.067
OLMo-2-7B	20	0.006	0.002	0.122
OLMo-2-7B	30	0.008	0.002	0.242
ModernBERT-large	3	0.044	0.044	0.089
ModernBERT-large	5	0.026	0.027	0.105
ModernBERT-large	10	0.017	0.013	0.150
ModernBERT-large	20	0.008	0.007	0.157
ModernBERT-large	30	0.005	0.004	0.143

Table 8: Per-target-token attention at the displaced slot in the last layer. “ $1/N$ ref.” is the budget-dilution baseline anchored at the smallest evaluated $N=3$; bold entries highlight the $N=30$ comparison. In three of four models the total block budget grows with N ; ModernBERT-large’s stays roughly flat.

Model	$N=1$	$N=3$	$N=5$	$N=10$	$N=20$	$N=30$	drop	95% CI
ModernBERT-large	0.622	0.585	0.636	0.727	0.744	0.740	0.058	[0.026, 0.097]
Qwen2.5-1.5B	0.356	0.432	0.450	0.405	0.406	0.414	0.136	[0.099, 0.171]
Qwen2.5-7B	0.446	0.518	0.589	0.621	0.621	0.596	0.116	[0.078, 0.165]
OLMo-2-7B	0.270	0.347	0.417	0.496	0.565	0.622	0.002	[0.000, 0.005]

Table 9: LAMA-style pilot: mean $P(\text{target})$ across 50 stratified words and three answer-prompt templates, for the `target_repeat` block at each N . Bold entries mark each model’s peak. Per-word drop is $(\text{peak}_w - P(N=30)_w)/\text{peak}_w$ with 95% bootstrap CI. Three of four models show a positive drop with CI excluding zero; OLMo-2-7B’s mean curve does not invert through $N=30$ and its CI lower bound rounds to zero at the reported precision (the unrounded 2.5% quantile is a small positive number, not strictly 0).

Verification. A verification script enumerates the expected output manifest (per-model target-word CSVs, displaced/ablation parquet tables, mechanism CSVs, figure files, and the compiled PDF) and reports per-file presence and size. The reference reproduction run passes the manifest in full.